Pentagon Pizza Index

Quantitative Proxy Strategy

Technical Documentation & Methodology

April 29, 2026

Abstract

This document describes a quantitative pipeline testing whether GDELT military tension spikes predict next-day defense and aerospace equity outperformance — a data-driven proxy for the Pentagon Pizza Index. The GDELT DOC 2.0 API is queried for three conflict themes (MILITARY_FORCE, TERROR, WMD); a daily tension score is constructed as the product of article volume and average tone negativity; a 28-day rolling z-score detects anomalous spikes. Three hypotheses are tested: VIX spike (H1), defense ETF outperformance (H2), and the simultaneous crisis trade (H3). Over 2017–2026, H2 and H3 pass at Fisher $p < 0.005$. ITA and PPA show statistically significant positive out-of-sample Sharpe in walk-forward validation (t-test $p < 0.03$). Initial paper trading was unprofitable due to position sizing and signal saturation; a revised implementation with Kelly cap, cooldown, and saturation filters addresses these issues.

> **Key finding:** *H1 (VIX) failed — markets rotated quietly into defense rather than panicking broadly. This is the signature of informed sector rotation, not fear-driven buying.*

1. The Hypothesis

1.1 The Pizza Index

The Pentagon Pizza Index is an informal market signal observed since at least the 1980s: delivery drivers and restaurant staff near the Pentagon notice unusual late-night orders before major military operations. Documented historical cases include the Kuwait invasion (1990), Desert Storm (1991), the bin Laden raid (2011), and various Middle East escalations in 2024. The signal is hyperlocal and not available as a historical dataset.

1.2 Quantitative Proxy via GDELT

GDELT (Global Database of Events, Language and Tone) monitors and aggregates global news coverage into structured event data. The DOC 2.0 API provides daily article counts and average tone by theme going back to 2017. Three conflict themes are queried:

Theme	Description	Role in signal
MILITARY_FORCE	Articles covering use or threat of military force	Primary — highest correlation with geopolitical events
TERROR	Terrorism events and counterterrorism responses	Secondary — captures asymmetric conflict
WMD	Weapons of mass destruction coverage	Tertiary — escalation signal; API most stable of the three

1.3 Three Hypotheses

Hypothesis	Prediction	Economic Logic
H1 — VIX spike	VIX rises ≥5% next day	Geopolitical shock creates broad market fear
H2 — Defense outperformance	ITA/XAR/PPA outperform SPY ≥0.5% next day	Defense sector benefits from conflict escalation expectations
H3 — Crisis trade	VIX up AND defense outperforms simultaneously	Informed rotation: fear + sector positioning

2. Signal Construction

2.1 Raw Tension Score

For each theme and each day, the raw tension contribution is:

```
tension_score = article_count × |avg_tone|
```

Article count captures how many news sources covered the event. Average tone captures how alarming the coverage was (GDELT tone ranges from −100 to +100; military conflict articles typically score −5 to −15). Multiplying the two weights large, alarming events more heavily than small or neutral ones. A skirmish covered by 500 outlets with tone −10 scores 5,000; a minor incident covered by 3 outlets with tone −5 scores 15.

Daily total = sum over all three themes. This is aggregated across all articles returned by the GDELT DOC 2.0 API for that day.

2.2 Pizza Z-Score

The raw tension score is right-skewed — most days are quiet, rare events are extremely large. Three transformations convert it to a tradeable signal:

```
# Step 1: Log-transform to compress extreme values
tension_log = log(1 + tension_raw)

# Step 2: 28-day rolling z-score
pizza_z = (tension_log - rolling_28d_mean) / rolling_28d_std

# Step 3: Spike flag
is_spike = pizza_z >= threshold   # default threshold = 2.0
```

The rolling z-score replicates the 'busier than usual vs recent baseline' logic used by pizzint.watch — a spike relative to the recent norm, not an absolute level. This is important because the baseline tension level changes over time (e.g., higher during the Russia-Ukraine war period than in peacetime). A day that would score z=3 in 2019 might only score z=1.5 during the sustained 2022 tension regime.

> **i** *The 28-day window is configurable via --window. Shorter windows (14 days) are more responsive but more volatile. Longer windows (60 days) are more stable but slower to detect regime changes.*

2.3 Data Fetching — Parallel Quarterly Batches

The GDELT v1.0 raw CSV server (403 Forbidden on automated clients) is not used. The GDELT DOC 2.0 API via gdelt-doc-api is queried instead, which is designed for programmatic access. Date range is split into 90-day chunks and all (theme, chunk) combinations are fetched in parallel via ThreadPoolExecutor:

```
# 3 themes × 38 chunks (2017-2026) = 114 API calls
# Run in parallel with 6 workers → ~90 seconds first run
# Cached to gdelt_tension_cache.parquet → instant on subsequent runs
```

3. Hypothesis Testing Framework

3.1 Test Battery

Each hypothesis is tested with multiple complementary methods. Using a single test would be fragile — Fisher's exact test is powerful for rare events but doesn't capture distribution shape; Mann-Whitney captures distribution differences but not relative frequency. Running all three provides convergent evidence.

Test	H0	Why used here
Fisher's exact test	P(outcome | spike) = P(outcome | no spike)	Best for rare events with unequal group sizes. Chi-squared is unreliable when spike days are 5% of observations.
Mann-Whitney U	Distributions of outcome are equal	Non-parametric — doesn't assume normality. Tests full distribution, not just a binary threshold.
Point-biserial correlation	No linear relationship between z-score and outcome	Measures effect size and direction. Confirms the relationship is monotonic.
Proportions z-test	Equal proportions of outcome in both groups	Provides a simple directional test for the binary outperformance flag.

3.2 Signal Alignment

GDELT data for day T is published at 6AM EST on day T+1. The strategy uses lag=1: signal observed on day T is used to trade on day T+1 (the next market open). This respects the data publication schedule and prevents look-ahead bias. GDELT calendar-day timestamps are forward-filled to align with market trading days (weekends and holidays are filled with Friday's signal).

> ⚠ *Never test with lag=0. GDELT day T data is not available until 6AM day T+1. Using lag=0 means trading on information you couldn't have had, overstating performance by 15-30% in simulations.*

3.3 Idiosyncratic Risk Check

Individual defense names (LMT, RTX, NOC, GD, LHX, BA) are tested for idiosyncratic dispersion on spike vs non-spike days. The coefficient of variation (CV = std/|mean|) ratio flags names where spike-day returns are far more dispersed than normal — indicating that company-specific events (earnings, contract announcements) are

coinciding with some GDELT spikes and contaminating the signal. LMT had CV ratio 6.60, NOC 1.64 — this is why the strategy uses ETFs rather than individual names.

4. Walk-Forward Validation

4.1 Design

A single train/test split is fragile — results depend heavily on which events land in which window. Walk-forward validation runs an expanding window:

```
for each fold:
  train = all data up to fold start
  optimise spike_z on train (min Fisher p across ETFs)
  test  = next 126 trading days (~6 months)
  record OOS Sharpe, win rate, n_spikes — no refitting on test
```

The z-threshold is the only hyperparameter optimised per fold. It is selected from candidates [1.2, 1.5, 1.8, 2.0, 2.2, 2.5] by minimising the mean Fisher p-value for H2 on training data. The selected threshold is then applied as-is to the test window.

4.2 Results Summary

ETF	Mean Sharpe	Std	Positive folds	Mean win rate	t-test p
ITA	+0.462	0.815	43% (6/14)	62.9%	0.0269 ✓
XAR	+0.319	1.063	43% (6/14)	65.6%	0.1410 ✗
PPA	+0.489	0.852	43% (6/14)	72.3%	0.0255 ✓

> **i** *43% positive folds is misleading — 7 of 14 folds had zero spikes (signal dormant). In the 7 active folds, 6 of 7 are positive (86%). The one negative fold (Jul 2021–Jan 2022) coincides with the Afghanistan withdrawal and Biden defense budget cuts — a regime where the economic mechanism broke down.*

4.3 Z-Threshold Stability

The walk-forward selected z=2.2 for folds 2–6 (early period with sparse spikes) and z=2.0 for all folds from 7 onward. The convergence to z=2.0 from fold 7 onward — where larger spike samples provided more reliable calibration — validates the default parameter choice.

5. Paper Trading Simulator

5.1 v1 Failure Analysis

The initial paper trader (v1) produced Sharpe = −0.706, total return −11.28%, and $13,497 in cost drag (13.5% of capital). Three structural problems caused this:

Problem	Root cause	Evidence
Kelly over-sizing	25% per ETF cap = 72% total capital deployed per spike	$13,497 cost drag on 429 trades; gross P&L ≈ 0 but net negative
No frequency filter	Strategy traded every spike including 85-spike cluster in 2025	Equity curve flat 2021–2024, then cliff-dive in 2025
No short selling available	SPY hedge requires shorting; not available on Robinhood	Long-only carries full market beta on every trade

5.2 v2 Fixes

Fix	Parameter	Rationale
Lower Kelly cap	kelly_cap=0.08 (8%)	Total capital deployed ~24% per spike. Cost drag reduced ~67% at same trade frequency.
Cooldown filter	cooldown_days=3	No trade within 3 days of last trade. Enforces 'rare event' assumption; prevents grinding when spikes cluster.
Saturation filter	saturation_max=8 / saturation_window =28	Skip if >8 spikes in last 28 days. Signal is no longer rare event at this frequency; rotation logic breaks down.
Robinhood cost model	0 commission, 1.5 bps bid-ask, 1.5 bps PFOF	Total ~6 bps round-trip vs ~7 bps. No SPY short (unavailable on Robinhood).

5.3 Kelly Fraction Calibration

Position size per ETF is determined by half-Kelly, calibrated on the training half of the data:

```
b = avg_win / avg_loss          # win/loss ratio on spike days
f* = (p*b - q) / b              # full Kelly fraction
f_half = f* * 0.5               # half-Kelly (standard conservative approach)
f_capped = min(f_half, kelly_cap)  # hard cap at 8%
```

Half-Kelly is used rather than full Kelly because full Kelly is theoretically optimal only with perfect parameter estimates. With 20–80 training observations for win rate and avg win/loss, the estimates carry significant uncertainty. Half-Kelly reduces position size to compensate, at the cost of some expected value.

6. Robinhood Cost Reality

Zero-commission brokers are not truly zero-cost. Understanding the actual cost structure matters for whether the strategy is viable.

Cost component	Traditional broker	Robinhood	Notes
Commission	~0.5–2 bps	0 bps	Genuinely free — no per-trade fee
Bid-ask spread	~1–2 bps	~1.5 bps	Pay half the spread on entry, half on exit — unavoidable market structure cost
PFOF slippage	~1 bps	~1.5 bps	Payment for order flow routes to market makers who fill at slightly worse prices
Short selling	Available	NOT available	SPY hedge is impossible — long-only only
Total round-trip	~7 bps	~6 bps	Small improvement, not the major driver of performance

> ⚠ *PFOF slippage is invisible — it shows up as fill prices slightly worse than the midpoint, not as a line-item fee. The 1.5 bps estimate is based on academic studies. Measure your actual fills vs the last trade price for the first 20 trades to calibrate this empirically.*

7. Code Guide

All code is in pizza_index.py (single file, ~1,167 lines). Key functions:

Function	Purpose
_date_chunks()	Splits date range into 90-day chunks for API batching
_fetch_chunk()	Fetches one (theme, date-range) from GDELT DOC 2.0. Retries 3× with exponential backoff.
fetch_gdelt_signal()	Orchestrates parallel fetching via ThreadPoolExecutor. Caches to parquet. Extends cache incrementally.
build_pizza_index()	log1p transform → 28d rolling z-score. Strips UTC timezone from GDELT index.
align_signal_to_market()	Reindexes GDELT (calendar days) to market (trading days) via ffill. Shifts by lag. Fixes tz-naive/tz-aware mismatch.
test_h1_vix()	Fisher exact + Mann-Whitney + point-biserial + proportions z-test for VIX response.
test_h2_defense()	Same test battery for defense ETF excess return vs SPY. Run per-ETF.
test_h3_combined()	Fisher exact on P(VIX up AND defense up | spike) vs base rate.
test_idiosyncratic()	CV ratio for individual defense names on spike vs non-spike days.
run_walk_forward()	Expanding window with per-fold z-threshold optimisation. Returns DataFrame of fold results.
_score_threshold()	Evaluates a candidate z-threshold on training data. Returns mean Fisher p (lower = better signal).
_kelly_fraction()	Half-Kelly position size. f* = (p*b-q)/b × 0.5, capped at max_f.
run_paper_trader()	Full paper trading loop with Kelly sizing, cooldown filter, saturation filter, cost deduction.

8. Quick Reference

8.1 Run Order

```
pip install gdelt-doc-api pandas numpy scipy statsmodels yfinance matplotlib

# Full run (first run ~90s for GDELT fetch, cached after)
python pizza_index.py

# Faster test (shorter window)
python pizza_index.py --start 2020-01-01

# Stricter spike threshold
python pizza_index.py --threshold 2.5
```

8.2 Output Files

File	Contents
pizza_index_results.png	9-panel diagnostic plot: tension series, z-score, VIX event study, boxplots, scatter, monthly frequency, ITA vs SPY
walk_forward_results.png	4-panel walk-forward: per-fold Sharpe, z* threshold, cumulative return, win rate
walk_forward_results.csv	Per-fold statistics: test window, best_z, n_spikes, Sharpe/win rate per ETF
paper_trader_results.png	6-panel paper trading: equity curve, drawdown, P&L scatter, z vs excess return, rolling Sharpe, cost drag
paper_trade_log.csv	Every individual trade: timestamp, ETF, z-score, Kelly fraction, notional, returns, P&L, skip reason
paper_equity_curve.csv	Daily equity, P&L, drawdown, spike flag
gdelt_tension_cache.parquet	Cached GDELT tension series — delete to force re-fetch

8.3 Interpretation Cheat-Sheet

If you see this…	It means…	Do this
Fisher p > 0.05 for all ETFs	Signal not significant in this period	Widen date range or lower threshold
Walk-forward folds mostly 0 spikes	Signal was dormant — not a code error	Check tension series; consider lower threshold

If you see this…	It means…	Do this
Cost drag > 5% of capital	Kelly fractions too large or too many trades	Lower kelly_cap or raise saturation_max
Sharpe negative in paper trader	Check skip counts: if saturation skips are high, 2025 cluster is dominating	Add --start 2017-01-01 --end 2024-01-01 to exclude cluster
GDELT API failures (warnings)	Theme not returning data for that date range	WMD theme is most stable; MILITARY_FORCE and TERROR have gaps
Timezone error on reindex	GDELT returns UTC, yfinance returns tz-naive	Already fixed in align_signal_to_market() and build_pizza_index()